

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

Marks

Q.1 Solve Any Two of the following.

- | | |
|---|---|
| A) Draw and explain Von Neumann architecture of computing? | 6 |
| B) Draw & explain top level view of computer system components? | 6 |
| C) Draw& explain functional view of computer system? | 6 |

Q.2 Solve Any Two of the following.

- | | |
|--|---|
| A) Explain various addressing modes with example? | 6 |
| B) Write a note on assembly level language and its instructions? | 6 |
| C) Compare RISC vs. CISC? | 6 |

Q.3 Solve Any Two of the following.

- | | |
|---|---|
| A) Explain the concept of SRAM and DRAM memory with typical memory cell structure? | 6 |
| B) A digital computer has a memory unit of 64KB and cache memory of 1KB word . Calculate the physical address as well as cache memory address for direct mapping? | 6 |
| C) Perform signed no. addition (i)(+13)+(8) (ii) (-13)+(8) (iii) (+13)+(-8) (iv)(-13)+(-8) | 6 |

Q.4 Solve Any Two of the following.

- | | |
|--|---|
| A) Compare between hardwired control unit & microprogrammed control unit | 6 |
| B) Write a note on microoperations? | 6 |
| C) Explain magnetic disk organization? | 6 |

Q.5 Solve Any Two of the following.

- | | |
|---|---|
| A) Draw and explain functioning of DMA mode with diagram? | 6 |
| B) Explain cache coherence with any solutions? | 6 |
| C) Explain multiprocessors & multicomputers organization. | 6 |

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch :Information Technology Semester :III

Subject Code & Name: BTITC303 & Computer Architecture and Organization

Max Marks: 60

Date: 06/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO's)	Marks
Q.1 Solve Any Two of the following.		12
A) What in general terms, is the distinction between computer organization and computer architecture?	CO1,2	6
B) What are the four main functions of a computer?	CO1,2	6
C) List and briefly define the main structural components of a computer.	CO1,2	6
Q.2 Solve Any Two of the following.		12
A) Write down 8086 assembly language program to add 16 bit number stored in memory location 3004H-3005H and 3006H-3007H.	CO1,2	6
B) Define registers. Enlist different registers and explain general registers in brief.	CO3	6
C) Convert the following decimal numbers in the binary form. a) 25.5 b) 10.625 c) 0.6875	CO2	6
Q.3 Solve Any Two of the following.		12
A) Explain secondary memory with its two types, applications and commonly used secondary memory devices.	CO3	6
B) Let us assume the even parity hamming code from the above example (111001101) is transmitted and the received code is (110001101). Now from the received code, detect and correct the error.	CO2,3	6
C) Write down difference between virtual memory and cache memory.	CO2,3	6
Q.4 Solve Any Two of the following.		12
A) Explain Control Unit and its Design in brief.	CO4	6
B) Write down and Explain the techniques which are based on the number of addresses utilized for micro-instruction sequencing.	CO4	6
C) What is Micro-programmed Control Unit?	CO3,4	6

Q. 5	Solve Any Two of the following.		12
A)	Explain Direct memory access controller in detail.	CO3,4	6
B)	Write short note on symmetric processor.	CO3	6
C)	Differentiate between Programmed I/O and Interrupt driven I/O.	CO3,4	6

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Information Technology

Semester : III

Subject Code & Name: BTITC303

Computer Architecture and Organization

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

Q. 1 Objective type questions. (Compulsory Question)

1 IR register is used for-----

- | | | | |
|---|---|--|--|
| a. to store most recently fetched instruction | b. to store most recently fetched operand | c. to store most recently fetched data | d. to store most recently fetched opcode |
|---|---|--|--|

2 RISC stands for-----

- | | | | |
|-------------------------------------|------------------------------------|--|--------------------------------------|
| a. Reduced instruction set computer | b. Reduced instruction set complex | c. Reduced instruction standard computer | d. Reduced introduction set computer |
|-------------------------------------|------------------------------------|--|--------------------------------------|

3 Choose the faster memory-----

- | | | | |
|--------|----------|--------------|-------------|
| a. RAM | b. Cache | c. Hard disk | d. Register |
|--------|----------|--------------|-------------|

4 Choose faster addressing mode-----

- | | | | |
|-------------|-----------|-------------|-------------|
| a. Implicit | b. Direct | c. Register | d. Indirect |
|-------------|-----------|-------------|-------------|

5 To represent the negative number -----presentation is used.

- | | | | |
|-------------------|-------------------|-------------------|-------------|
| a. sign magnitude | b. 1'S Complement | c. 2'S Complement | d. both b&c |
|-------------------|-------------------|-------------------|-------------|

(Level Marks
I/CO)

12

1

1

1

1

1

6	Greater Storage capacity of ----- external memory supports.	1
	a. CD b. DVD c. CD-ROM d. RAID	
7	Today's computer system supports ----- IO technique.	1
	a. Programmed IO b. Interrupt-driven IO c. DMA Controller d. Both a & c	
8	Physically address is represented is with two bits -----	1
	a. S+W b. W+S c. R+W d. TAG+ S	
9	Larger unit to map the memory -----	1
	a. Blocks b. words c. bits d. bytes	
10	Cache memory logically divided into -----	1
	a. Tag b. Blocks c. byte d. Lines	
11	-----Assembler translates assembly code into machine code.	1
	a. TASAM b. MASAM c. Both a & b d. none	
12	Cache coherence can be handled with -----	1
	a. snoopy protocol b. Directory protocol c. MESI protocol d. all a, b & c	
Q. 2	Solve the following.	12
A)	Explain (Top level view) major components of computer system components present on the motherboard with neat diagram?	6
B)	Define the terms - Computer, Organization, Architecture, Structure, and Function?	6
Q.3	Solve the following.	12
A)	Explain functional view of computer system components with neat diagram?	6
B)	Explain machine instruction with its characteristics and types?	6
		12

Q.4 Solve Any Two of the following.

- A) Write a note on internal and external memory organization? 6
- B) Explain error detection and correction mechanism with example? 6
- C) Perform binary addition with 8 bit register (i) $(+55) + (27)$ (ii) $(+55) + (-27)$ (iii) $(-55) + (27)$ (iv) $(-55) + (-27)$ for negative no. use 2's complement representation? 6

Q.5 Solve Any Two of the following.

- A) Write a note on virtual memory system? 6
- B) Write a note on MESI Protocol? 6
- C) Write a note on Flynn's Classification? 6

Q.6 Solve Any Two of the following.

- A) Give the difference between micro-programmed and hardwired control unit 6
- B) Explain working of DMA Controller with neat diagram? 6
- C) Explain instruction pipelining with its issues? 6

*** End ***